# Turgut Aghashov

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**Research Interests:** Operating Systems Security | Kernel Analysis | Automated Vulnerability Discovery

## EDUCATION

**University of Maryland** College Park, MD, USA
Professional Master of Engineering Aug 2024 - Dec 2025

- **Specialization:** Cybersecurity Engineering
- **Relevant coursework:** Hacking of C programs and Unix binaries, Networks and Protocols, Cloud Computing, Advanced Hacking of Linux, Network Security, Business of Engineering, Reverse Engineering, Fundamentals of AI and Deep Learning, Intro to Secure Coding, Secure Operating Systems.
- **GPA:** 3.88/4.0

**Baku Higher Oil School** Baku, Azerbaijan
Bachelor of Engineering with Honors Sep 2019 - Jul 2024

- **Specialization:** Information Security
- **Relevant coursework:** Web Programming and Security, Mobile Programming and Security, Database Administration and Security, Network Security, Data Structures and Algorithms.
- Delivered Red Team training sessions for junior students in Cyber Club.
- **GPA:** 95.57/100

## RESEARCH EXPERIENCE

**Internship Research - RefProbe: Hunting Kernel Lifetime Bugs with eBPF**
Bureau Veritas Cybersecurity North America Jul 2025 - Aug 2025

- Built an eBPF-based tool to detect Linux kernel lifetime bugs, including use-after-free, double-free.
- Developed novel trace analysis algorithms that correlate allocation and deallocation events across kernel subsystems to identify inconsistent reference patterns and dangling pointer access.
- Engineered test kernel modules to inject controlled bugs and measure detection coverage and false positives.
- Achieved 87% detection rate on injected lifetime bugs with <5% false positives, demonstrating practical applicability for kernel development workflows.

**Undergraduate Thesis Research - Mobile Security Ecosystem and Penetration Testing**
Baku Higher Oil School Feb 2024 - May 2024

- Performed security assessment of Android banking application using static and dynamic analysis (MobSF, Frida, Drozer, Burp Suite) to identify vulnerabilities across application, OS, and network layers.
- Mapped findings to OWASP Mobile Top 10 and proposed concrete remediation steps, demonstrating how systematic pentesting hardens real-world mobile apps.
- Demonstrated exploit chains combining client-side and server-side vulnerabilities that could expose sensitive financial data for millions of users.

## PUBLICATIONS

**Building an Effective Security Operation Center with Open-Source Tools. T. Aghashov** and N. Najafli
4th International Scientific Conference of Students and Young Researchers, Baku Higher Oil School Apr 2023 - May 2023

- Designed reference SOC architecture integrating open-source SIEM (ELK Stack), IDS (Suricata), and log analysis tools tailored for resource-constrained small and medium-sized enterprises.
- Conducted comparative analysis of open-source versus commercial SOC solutions, demonstrating 60% cost reduction while maintaining comparable detection coverage for common attack patterns.
- Produced deployment guidelines and configuration templates enabling practical implementation for organizations with limited security budgets.

## COURSE PROJECTS

**ENPM 693 - Man-in-the-Middle Attacks on the Southbound Interface in Software-Defined Networks**

- Built a CORE + Ryu-based SDN testbed to emulate controller-switch communication over OpenFlow.
- Implemented ARP-spoofing and traffic-relay attacks to intercept and modify southbound traffic in real time.
- Evaluated detection and mitigation strategies using flow-rule validation, anomaly monitoring, and controller-side checks.

**ENPM 818N - Secure AWS Cloud Architecture**

- Designed multi-AZ VPC architecture with defense-in-depth security controls including AWS WAF, Shield, and VPC flow logs, demonstrating practical application of cloud security principles.

- Optimized static asset delivery via CloudFront + S3, reducing attack surface while improving performance and reliability.

**ENPM 680 - UMDiet: Secure Go Web Application**
- Implemented Go-based diet-tracking and planning app with authenticated endpoints and server-side validation.
- Used modular APIs, Git workflows, and reviews to mirror secure development practices.

## INDUSTRY EXPERIENCE

**Bureau Veritas Cybersecurity North America** — Seattle, WA, USA
Security Engineering Intern — Jun 2025 - Aug 2025
- Identified 150+ security flaws through advanced web & cloud penetration testing lab environments.
- Assessed 3 enterprise Web/Infrastructure systems, uncovering critical misconfigurations and producing remediation reports.

**Cyberpoint LLC** — Baku, Azerbaijan
Leading Penetration Testing Specialist — Dec 2023 - Aug 2024
- Executed black-, gray- and white-box penetration tests for 5 banks, 2 state agencies, and 2 telecoms.
- Exposed 100+ vulnerabilities, including OWASP Top 10, preventing potential data exposure risk for millions of users.

**Prosol CJSC** — Baku, Azerbaijan
Junior Cyber Research and Assessment Specialist — Aug 2022 - Nov 2023
- Conducted web, mobile, and infrastructure penetration tests across 4 banks and 2 enterprises.
- Traced and analyzed 5 phishing campaigns targeting government agencies, recommending defensive measures.

## CERTIFICATIONS

OffSec Web Expert (**OSWE**) by OffSec — Oct 2024
eLearnSecurity Mobile Application Penetration Tester (**eMAPT**) by INE — Feb 2024
OffSec Certified Professional (**OSCP**) by OffSec — Nov 2023
Certified Red Team Operator (**CRTO**) by Zero-Point Security — Jun 2023

## AWARDS

**1st Place at the Recyclaton - 2022 Hackathon** — Baku, Azerbaijan
*The Coca-Cola Foundation and ASAN Services* — Dec 2022
- Secured first place with "Replay" team, awarded 6,000 AZN (~ $3,500) as seed funding for the project idea.

**Presidential Scholarship Recipient** — Baku, Azerbaijan
*Government of Azerbaijan* — Sep 2019
- Awarded a full Presidential Scholarship for scoring 689.4/700 on the national university entrance examination in Azerbaijan (ranking 4th nationwide).

**Golden Medal for Excellency** — Baku, Azerbaijan
*The Ministry of Science and Education of the Republic of Azerbaijan* — Jun 2019

**Bronze Medal at Republican Math Olympiads** — Baku, Azerbaijan
*The Ministry of Science and Education of the Republic of Azerbaijan* — May 2019

## ACTIVITIES

**Math Coach - Math Success Program, UMD** — Oct 2025 - Dec 2025
- Tutored MATH120/140/113/141 students in precalculus and calculus problem-solving.
- Simplified concepts and exam-style questions to build confidence and knowledge.

**Cryptography Teaching Assistant - Azerbaijan Cyber Security Organizations Association** — Jun 2023 - Jul 2023
- Tutored 30 students in understanding key concepts and practical applications in cryptography.

## SKILLS

**Programming Languages:** Python, C, C++, Go, Bash, JavaScript, Java, PHP, SQL
**Security Tools & Frameworks:** Burp Suite, OWASP ZAP, Metasploit, Wireshark, MobSF, Frida, JADX, Nmap, Ghidra, GDB, Pwntools, IDA Pro, Volatility
**Cybersecurity Specializations:** Binary Exploitation, Reverse Engineering, Penetration Testing (Web, Mobile, Network), Secure Code Review, DevSecOps
**Cloud Technologies:** AWS (EC2, VPC, RDS, S3, ELB, ASG, CloudFormation, Route53, Lambda, CloudWatch), GCP, Azure
**Open-Source Tools:** Burp Suite, OWASP ZAP, Metasploit, Wireshark, MobSF, Frida, JADX, Nmap, Ghidra, GDB, Pwntools
**CI/CD Tools:** Git, Github Actions, Docker, Ansible, Kubernetes